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NUMERICAL SIMULATION AND EXPERIMENTAL STUDY OF DRAG-REDUCING SURFACE OF A REAL SHARK SKIN*

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Abstract: It is well known that shark skin surface can effectively inhabit the occurrence of turbulence and reduce the wall friction, but in order to understand the mechanism of drag reduction, one has to solve the problem of the turbulent flow on grooved-scale surface, and in that respect, the direct numerical simulation is an important tool. In this article, based on the real biological shark skin, the model of real shark skin is built through high-accurate scanning and data processing. The turbulent flow on a real shark skin is comprehensively simulated, and based on the simulation, the drag reduction mechanism is discussed. In addition, in order to validate the drag-reducing effect of shark skin surface, actual experiments were carried out in water tunnel, and the experimental results are approximately consistent with the numerical simulation.

Key words: numerical simulation, experimental study, biomimetic surface, shark skin, drag reduction mechanism

Introduction

Through millions of years' competition for survival of the fittest, a creature has formed his own unique functional surface, such as for drag reduction, wear resistance, self-cleaning and so on. Shark is one of the fastest swimming animals in the ocean, with many micro-grooves on the surface of shark skin scales. In the 1970s, Michael in NASA Langley Research Center found that the ribbed surfaces can effectively reduce the wall friction[1], which is greatly out of the traditional way of thoughts, and which has also opened up new ways of drag-reducing technology for mankind.

The direct numerical simulation is an important tool to study turbulent flow, and many studies are closely related with fluid engineering[2-11]. The mechanism of grooved drag-reducing surface has been one of important topics[12-16]. However, the simulations so far are mainly confined to simple straight grooves, such as triangle groove, semicircular groove and other simple straight grooves[12-16], and to the authors' knowledge, the flow simulation over a real shark skin surface has not been reported, additionally, the influence of attack angle is not considered. Moreover, in order to verify the drag reduction effect of shark skin, actual experiments are carried out in water tunnel.

1. Building model of shark skin

Shark skin is composed of many small scales, and the micro morphology of surface is shown in Fig.1. The extension direction of scale is approximately parallel with its swimming direction, the groove tips can stick out of the viscous sublayer and inhabit the occurrence of turbulence, so the wall friction can be reduced.

To study the drag reduction mechanism of a real shark skin surface, a model is built in the following steps: (1) high-accurate scanning of a single biological scale, (2) data analysis and processing, (3) building of the three-dimensional model of single scale and shark skin in a large area, (4) building of the computational domain, (5) building of CFD model and numerical

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simulation. Because all relevant information of the model comes from a real biological shark skin prototype, so the simulation is more accurate and reliable than that of ordinary straight grooves.

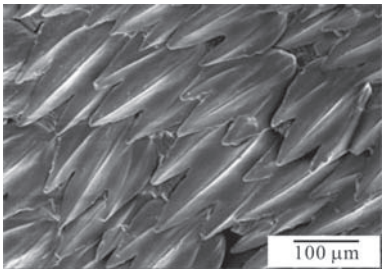

Fig.1 SEM image of shark skin

1.1 *Pretreatment and scanning of biological shark skin*

In the process of building the model, to achieve good optical reflection and electric conduction, the shark skin goes through the following five steps: cleaning, chemical fixation, re-cleaning, dehydration, drying[17,18]. To obtain sufficient and precise data, the super-high accurate scanning method is adopted, and the high precise instruments include Phase Shift MicroXAM-3D, with RMS repeatability of 1 nm, the minimum of vertical scanning resolution of 0.1nm and the calibration accuracy less than 0.1%. By high accurate scanning, the three-dimensional image of the single scale is obtained as shown in Fig.2.

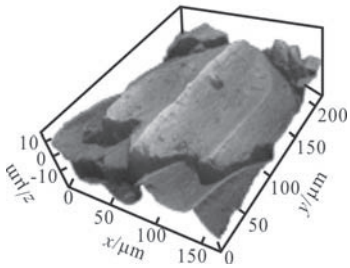

Fig.2 Three-dimensional image of a single scale

1.2 *Building model of real shark skin*

Building the model of a real shark skin is one of the most important steps in the process, and the model should be exactly consistent with the biological prototype. The two-dimensional image of the single scale surface and the corresponding cross-section curve of signed lines are shown in Fig.3. With the relevant data being imported into Solidworks software, by the steps of lofting, clipping and so on, the three-dimensional models of the single scale and the shark skin surface with many scales are built, as shown in Fig.4 and Fig.5.

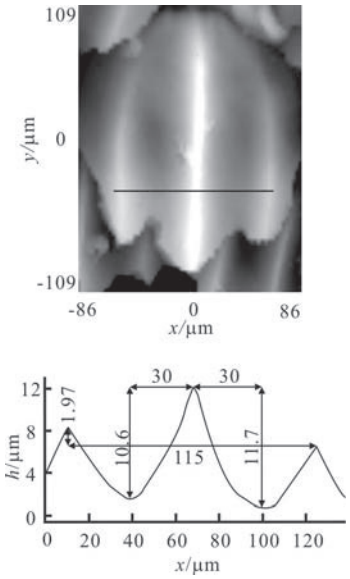

Fig.3 Biological shark skin scale and corresponding cross-section curve

Fig.4 Single scale of shark skin

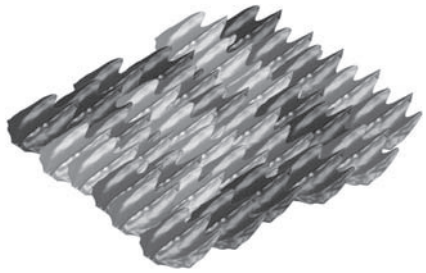

Fig.5 Shark skin surface with many scales

2. CFD pre-processing

Because the shark skin scale surface is so complicated, it is very difficult to paste different surfaces with no gaps, and the continuity of the shark skin surface and the computational domain cannot be ensured. Therefore, the real shark skin surface should be simplified in a reasonable way. In addition, the

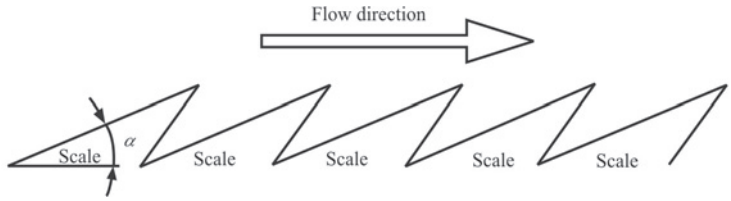

Fig.6 Attack angle of shark skin scale

longitudinal section of the scale surface is not parallel to the flow direction, but in a particular attack angle, as shown in Fig.6. Based on the built scale model and the actual attack angle (it is about 10° - 30°, the attack angle of all scales is assumed to be 15° in this article), the continuous real shark skin surface model and the computational model are obtained as shown in Fig.7 and Fig.8.

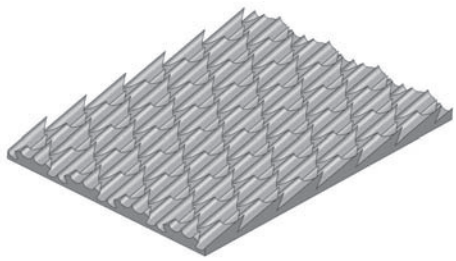

Fig.7 Continuous real shark skin surface

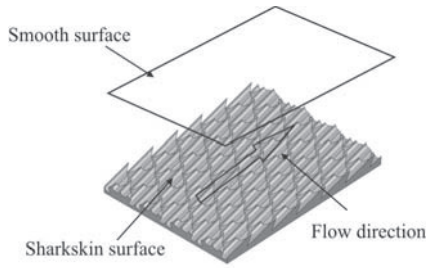

Fig.8 Computational model

To ensure a high accuracy of computation and to reduce the total number of grids, the distribution should comply with the following principles[12-14]:

(1) In order to capture the features of the flow field near the walls, the grids near the upper wall and the lower wall should be very refined. (2) Limited by the capacity of computer, the total number of grids in the computational domain should not be greater than 1 600 000, so the grids should be sparse in the region far from the walls.

The grids of the model and the enlarged grids near the shark skin scale surface and the smooth surface are shown in Fig.9, with the width, height and length of the computational domain in the ratio of 10:4:15, and the height of the domain being about 20

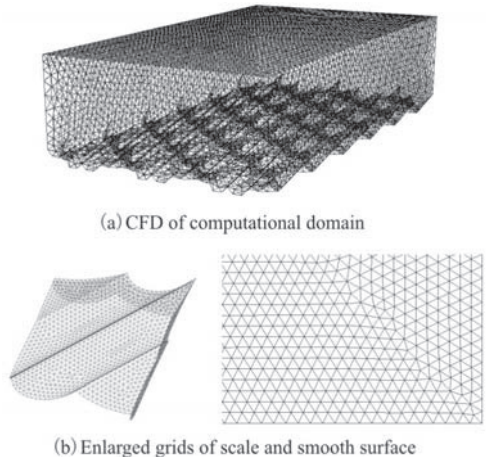

Fig.9 Distribution of grid

times of the height of the scale, therefore, the flows near the upper and lower walls will not be disturbed, and the accuracy of the simulation can be ensured. On the vertical direction, the height is divided into 40 line-grids in the ratio of 1.25:1. The parameters in meshing the whole volume include Elements-Tet/Hybrid, Type-TGrid, and Spacing Interval Size-5. The total number of grids is about 1 470 000, the maximum and minimum of grid volume are, respectively, 4.496124×10^{-16} m^3 and 3.574049×10^{-20} m^3.

In the course of simulation, the RNG $k-\varepsilon$ turbulence model is adopted, with the Enhanced Wall Treatment near the wall to exactly simulate the flow field near the tiny grooves, and with other options set as default. In addition, the boundary conditions are treated as follows[12-14]:

(1) Because of the fully developed turbulent, the flow inlet and flow outlet of the computational flow field should be set as in a periodical boundary condition.

(2) Among the span wise of the flow field, the flow is not constrained, in order to avoid other disturbance of the flow field, the boundary along the span wise of the computational flow field is set as in a symmetry condition.

(3) The upper smooth wall and the lower real shark skin surface wall are set as in no-slip and

impenetrable boundary condition.

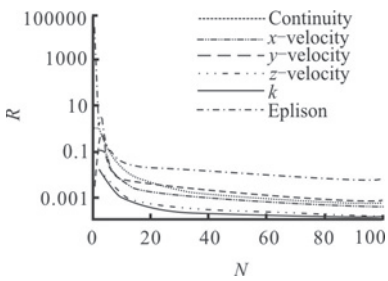

Fig.10 Residual curves of numerical simulation

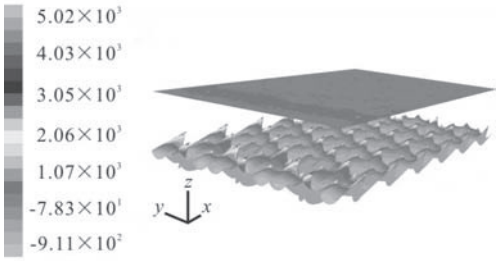

Fig.11 Contours of *X*-wall sheer stress

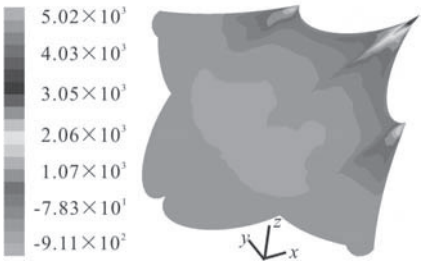

Fig.12 Contours of *X*-wall sheer stress on a single scale

3. Flow field analysis on real shark skin surface

3.1 *Analysis of sheer stress*

When the average velocity in the flow field is 6 m/s, the Reynolds number exceeds the critical value, and the flow comes into a fully developed turbulent state, with the depth of viscous sublayer being only about 10 μm, therefore, the tips of scales would stick out the layer, and the grooved surface plays a role of inhabiting the occurrence of turbulence and in turn, reducing the drag. In the process of numerical simulation, the number of iterations is set as 100, the curves of residuals are shown in Fig.10, the contours of sheer stress in the flow direction over real shark skin surface and smooth surface are shown in Fig.11. In order to further analyze the sheer stress, the contours of sheer stress on a single scale are also obtained, as shown in Fig.12.

According to the above figures, the sheer stress on the smooth surface is almost uniform, about 408 Pa - 738 Pa, but on the single real shark skin scale it takes a value of about −580 Pa - 4 700 Pa on the flow direction, with the maximum on the tip, which is larger than that on the smooth surface, the minimum on the valley, in the direction or even opposite to the flow direction, so much less than that on the smooth surface. The integral of the sheer stress over the real shark skin surface is also less than that on the smooth surface, so the real shark skin shows a drag reduction effect.

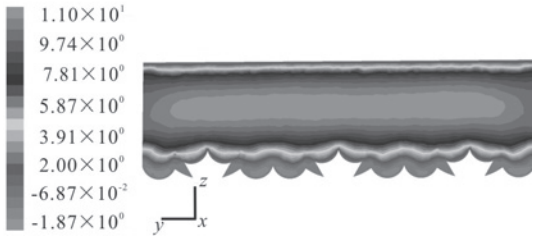

Fig.13 Contours of *X*-velocity on a particular cross-section

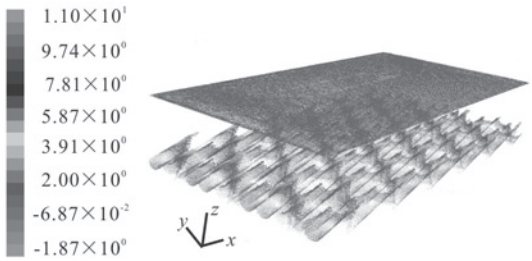

Fig.14 *X*-velocity vector on a smooth surface and the real shark skin

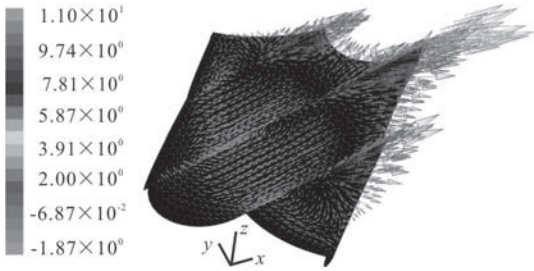

Fig.15 *X*-velocity vector on a single scale

3.2 *Analysis of velocity field*

The velocity takes a different form in different parts of the flow field. Figure 13 shows the velocity contours on a particular cross-section of the flow field, and Fig.14 shows the velocity vector on a smooth surface and the real shark skin surface in the flow direction, and the velocity on a single scale is shown in Fig.15.

As shown in Fig.13, when the average velocity is 6 m/s, influenced by the upper and lower walls, in the region far from the wall, the velocity is larger, and the maximum and the minimum of velocity are greater than 10 m/s and less than zero, respectively. More-

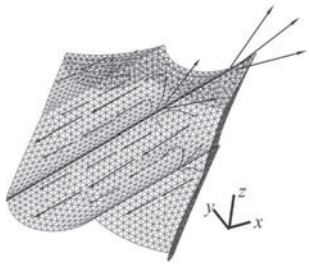

Fig.16 Diagram of back flow on scale surface

over, the velocity in the viscous sublayer is varying very rapidly, with a low velocity and small friction near the wall. For the single scale, the velocity on the tip is very large, and the velocity on the valley is very small, even takes a negative value, as backflows, which leads to the reduction of the wall friction. The diagram of back flows on the scale surface is shown in Fig.16.

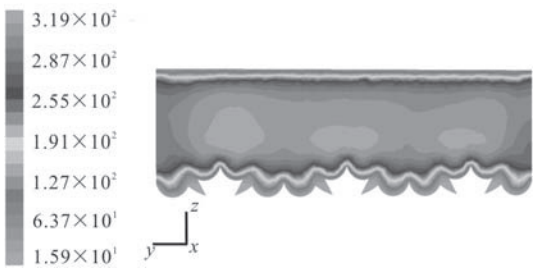

Fig.17 Contours of turbulence intensity on cross-section

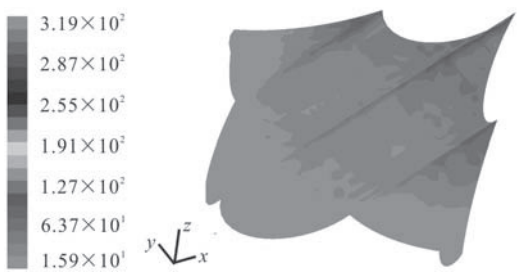

Fig.18 Contours of turbulence intensity on a single scale

3.3 *Analysis of turbulence intensity*

The turbulence intensity may directly affect the flow state and influence the wall friction. The contours of turbulence intensity on a particular cross-section and the single scale are shown in Fig.17 and Fig.18. As shown in the figures, the turbulence intensity in the center of the flow field is larger than that in other regions, and the intensity near the smooth surface is larger than that near the valley of the scales, and for the single scale, as shown in Fig.18, the intensity on the tips is larger than that on the valley. Theoretically speaking, the greater the turbulence intensity, the larger the wall friction will be, therefore, the wall friction on the shark skin is less than that on a smooth wall.

When the average velocity of the flow field is 6 m/s, the maximum of the turbulence intensity on the cross-section is in the middle of the cross-section, about 302.65414% - 318.58325%, and the turbulence intensity attached to the upper wall and lower wall are 31.85948% - 47.788574% and 0.001281% - 15.9303%, respectively. Moreover, the turbulence intensity on the tip is much larger than that on the valley of the scale.

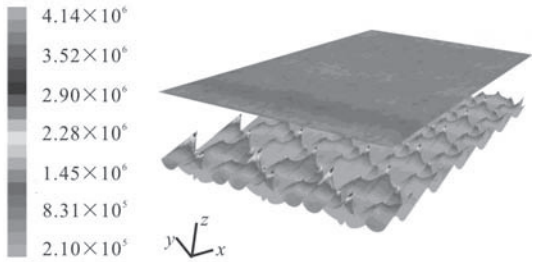

Fig.19 Contours of vorticity magnitude on smooth surface and shark skin

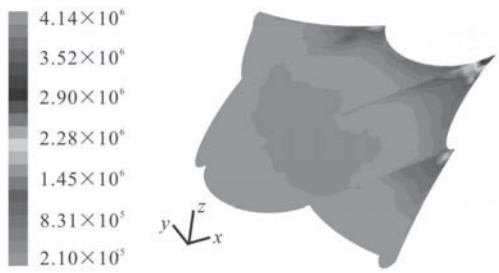

Fig.20 Contours of vorticity magnitude on single scale

3.4 *Analysis of vortex*

When the average velocity of the flow field is 6 m/s, the contours of vorticity of magnitude on the shark skin and the smooth wall are shown in Fig.19 and Fig.20. The intensity takes value mostly in the range of 624 190.31 l/s - 831 259.19 1/s on the smooth surface, and on the shark skin, the maximum on the tip is about 3 523 154.5 l/s - 3 730 223.5 1/s, the minimum on the valley is about 2 983.7073 l/s - 210 052.58 1/s. The integral vorticity magnitude over the shark skin surface and the smooth surface is 0.3941 and 0.4653, respectively. Therefore, it is concluded that the spreading movement of the vorticity is restricted by the grooves of the shark skin, which leads to drag reduction.

3.5 *Drag-reducing efficiency of real shark skin*

In the computational field, when the average velocity exceeds 5 m/s, the turbulence can take place and the shark skin surface has the effect of drag reduction. The different mass flows of water are considered in accordance with the different average

velocities, the drag-reducing efficiency of the real shark skin at different velocities can be obtained, as shown in Table 1.

Table 1 Drag-reducing efficiency of real shark skin

Average velocity (m/s)	Drag-reducing efficiency	Average velocity (m/s)	Drag-reducing efficiency
5.1	7.09%	5.6	9.36%
5.2	7.53%	5.7	10.21%
5.3	7.69%	5.8	11.69%
5.4	8.08%	5.9	12.53%
5.5	8.28%	6.0	13.63 %

4. Experiments on shark skin

4.1 *Drag-reducing experiment of shark skin*

In order to validate the drag reduction effect of the real shark skin surface, the drag-reducing experiment is conducted in vacuole water tunnel (China Ship Scientific Research Center (CSSRC)). The parameters used in the test system are as follows:

(1) The test section in vacuole water tunnel is 3.2 m in length and 0.8 m in diameter.

(2) The flow velocity in test section is adjustable between 0 m/s - 20 m/s.

(3) The central pressure of the test section can be varied between 8 kPa - 400 kPa.

(4) The vacuole index is 0.15.

The test instruments also include a resistance sensor of 300 N in measurement range and 0.1 N in resolution, two SA55 DC amplifiers, NI-PCI4472 digital sampling panel with sampling frequency of 200 KHZ, and IBM-Piv computer.

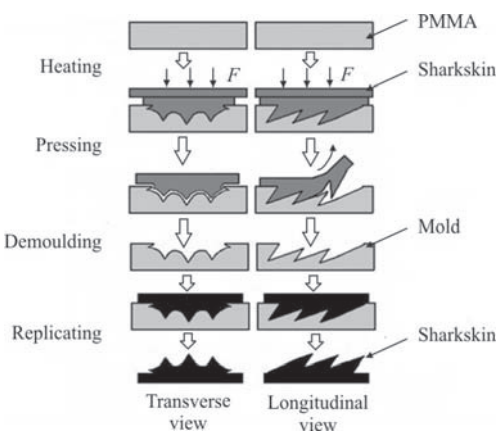

Fig.21 Fabrication of real shark skin[17,18]

The testing model with hollow elliptical sphere at front and hollow cylinder at posterior segments are made of aluminum alloy LY12. The size of the test model with the length of the cylinder section of 500 mm and external diameter of 90 mm is determined according to the size of strain gauge balances of the test system. The external diameter of hollow elliptical sphere is 94 mm to guarantee outside surface smoothness after pasting the test skin sample. The real shark skin surface can be fabricated by bio-complicated method[17,18], as shown in Fig.21. In the process of pasting, the direction of micro-grooves of skins should be consistent with the direction of flow. The test model and installation photos are shown in Fig.22.

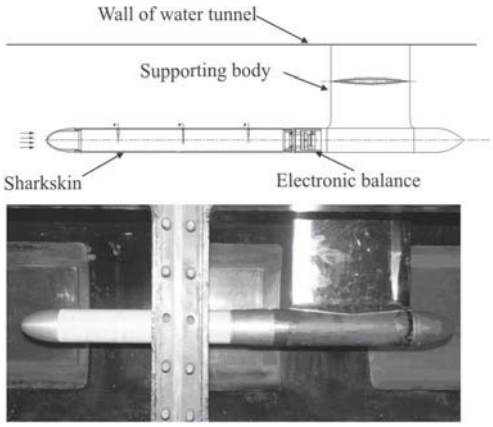

Fig.22 Test model and its installation photos

The experiments are conducted according to the requirements for vacuole water tunnel test (Q/702J0301-2008). The water temperature is 28℃ and the test system is degassed for more than 1 h before testing. The water velocity can vary from 2 m/s to 10 m/s. The water velocity of each test continuously rises until the skins are washed off. The drag-reducing efficiency of the real shark skin can be determined as shown in Table 2.

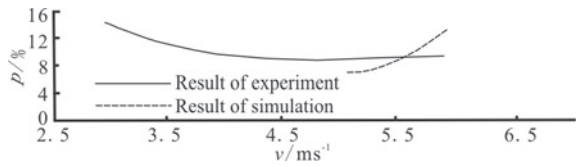

Fig.23 Drag-reducing efficiency of the real shark skin

4.2 *Comparison of results between simulation and experiment*

The drag-reducing efficiency of the real shark skin obtaining from numerical simulation and actual experiment is shown in Fig.23.

As shown in Fig.23, the results are approximately consistent, which verifies the accuracy and reliability of numerical simulation and experiment in some extent. The following factors would contribute to the deviation: the actual testing samples are smooth skins

Table 2 Experimental results of shark skin

Velocity (m/s)	Smooth surface (N)	Real shark skin surface (N)	Drag-reducing efficiency	Velocity (m/s)	Smooth surface (N)	Real shark skin surface (N)	Drag-reducing efficiency
3.3	2.42	2.11	12.81%	4.5	5.76	5.22	9.375%
3.6	3.11	2.76	11.25%	5.0	7.61	6.91	9.198%
3.9	3.89	3.50	10.03%	5.5	9.75	8.90	8.72%
4.2	4.78	4.32	9.62%	6.0	12.15	10.99	9.55%

Table 3 Component force on real shark skin surface and smooth surface

Zonename	Pressure force (N)	Viscous force (N)	Total force (N)
Shark skin surface	0.0002492357916	0.0001170265777	0.0003662623693
Smooth surface	0	0.00039931147	0.00039931147

and real shark skins made of silicon rubber, but the models are rigid in the numerical simulation, therefore the results may be influenced by the difference of hardness; in addition, constrained by the size of the whole shark skin surface, the testing samples of smooth skin and fabricated shark skin are both prepared by pasting several skins, so the seams on the samples may affect the drag-reducing effect.

5. Analysis of drag reduction mechanism of shark skin

Table 3 shows the component force on the smooth surface and shark skin in the flow direction when the average velocity is 5.5 m/s. Based on the table, the wall friction of the shark skin is composed of the pressure force and viscous force, which are produced and influenced by the attack angel and the micro-grooves of the scale.

From the data of the shark skin and smooth surface, the following conclusions can be drawn:

(1) Influenced by the micro-grooves and attack angle of scales, the turbulence intensity is reduced, and with “back flow” on the scale surface, the viscous force on the shark skin is much decreased as compared with that on the smooth surface.

(2) Due to the existence of attack angle of scales, the pressure force on the shark skin is generated. However, the total force (pressure force plus viscous force) on the shark skin surface is less than that on the smooth surface. Therefore, the reduction of turbulence intensity for drag reduction more than offsets the generated pressure force.

Moreover, when a shark is swimming very fast, it can release mucus, which can play the role of lubricating[19]. This is another crucial factor in decreasing the resistance.

6. Conclusions

In this article, by direct numerical simulation and actual experiment, the drag-reducing efficiency and mechanism of shark skin are investigated. The main conclusions are as follows:

(1) The method of high-accurate scanning of the single biological scale is implemented, and the three-dimensional real shark skin model is exactly built, as is more consistent with the biological prototype, so convincing and reliable simulation results can be obtained.

(2) In the course of simulation, the attack angel of scale is taken into consideration, which can lead to the occurrence of tiny backflow on the valley of the scale, and the micro-grooves and attack angle of the scale are the main contributions for drag reduction.

(3) The actual experiment is performed in the water tunnel, and the results are approximately consistent with the simulation, which verified the accuracy of simulation and experiment in some extent.

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